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Github Link: https://github.com/BasithZhang/DM2026-Assignment-1.git

HOMEWORK 1

1. Introduction

This assignment investigates the behavior of supervised learning models, including Linear Regression, Logistic Regression, and a real-world classification task. The primary objective is to analyze how hyperparameters such as learning rate, number of iterations, and regularization strength influence model performance.

All models are implemented under a unified LinearModel framework, where different learning tasks are configured through loss functions, gradients, and activation functions. This unified design allows consistent comparison across regression and classification problems. To ensure reproducibility, all experiments are conducted using fixed random seeds as required in the assignment.

2. Task 1: Linear Regression

2.1 Experimental Setup

We apply Linear Regression on four datasets (A–D). For each dataset, we evaluate three learning rates:

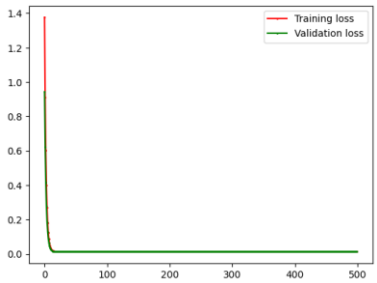
- 0.1
- 0.01
- 0.001

The model is trained using Mean Squared Error (MSE), and evaluated using:

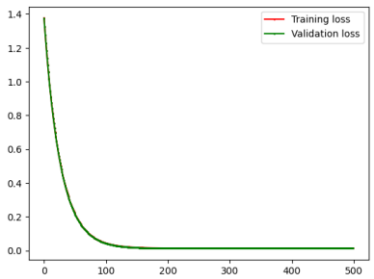
- MSE
- MAE
- RMSE
- R^2

2.2 Dataset A

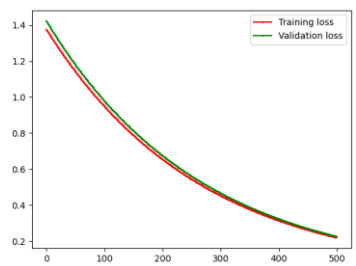
To analyze the effect of learning rate on optimization behavior, we visualize the training and validation loss curves.




```
=== Linear Regression Evaluation | Dataset A | lr=0.1 ===
MSE       : 0.0134
MAE       : 0.1010
RMSE      : 0.1156
R-squared : 0.5690
```




```
=== Linear Regression Evaluation | Dataset A | lr=0.01 ===
MSE       : 0.0134
MAE       : 0.1009
RMSE      : 0.1156
R-squared : 0.5690
```




```
=== Linear Regression Evaluation | Dataset A | lr=0.001 ===
MSE       : 0.2201
MAE       : 0.3784
RMSE      : 0.4692
R-squared : -6.1039
```

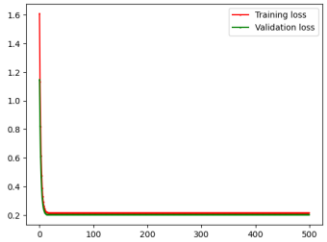
As shown in these pictures, the learning rate of 0.1 reduces the loss rapidly during early iterations, indicating fast convergence. However, small oscillations are observed, suggesting that the update steps are relatively large and may overshoot the optimal region.

In contrast, the learning rate of 0.001 produces a smooth but slow convergence, indicating that updates are too small to efficiently reach the optimum within the given iterations. The learning rate of 0.01 provides a balance between speed and stability. It converges consistently without significant fluctuations. The evaluation results support this observation, where lr = 0.01 typically achieves lower MSE and higher R^2 compared to other settings.

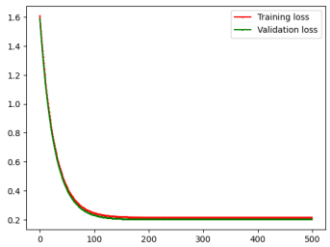
2.3 Dataset B, C, and D

To further validate the consistency of learning rate behavior, we repeat the same experiment on Datasets B, C, and D.

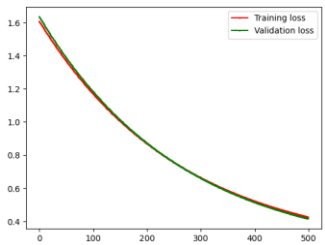
a. Dataset B




```
=== Linear Regression Evaluation | Dataset B | lr=0.1 ===
MSE       : 0.2137
MAE       : 0.4038
RMSE      : 0.4623
R-squared : 0.0448
```

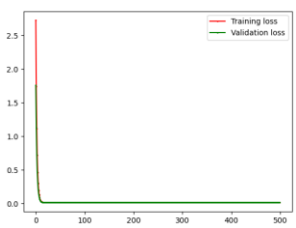



```
=== Linear Regression Evaluation | Dataset B | lr=0.01 ===
MSE       : 0.2137
MAE       : 0.4038
RMSE      : 0.4623
R-squared : 0.0448
```

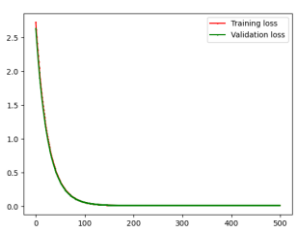



```
=== Linear Regression Evaluation | Dataset B | lr=0.001 ===
MSE       : 0.3920
MAE       : 0.5172
RMSE      : 0.6261
R-squared : -0.7521
```

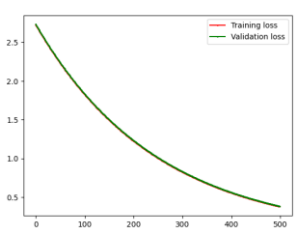
b. Dataset C




```
=== Linear Regression Evaluation | Dataset C | lr=0.1 ===
MSE       : 0.0132
MAE       : 0.0987
RMSE      : 0.1148
R-squared : 0.9954
```

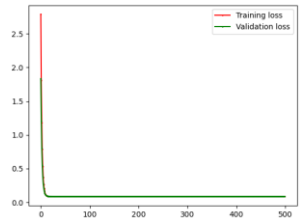



```
=== Linear Regression Evaluation | Dataset C | lr=0.01 ===
MSE       : 0.0132
MAE       : 0.0987
RMSE      : 0.1148
R-squared : 0.9954
```

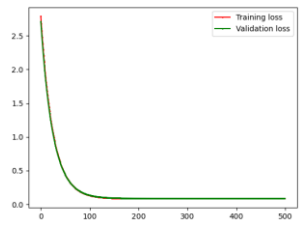



```
=== Linear Regression Evaluation | Dataset C | lr=0.001 ===
MSE       : 0.3796
MAE       : 0.5018
RMSE      : 0.6161
R-squared : 0.8662
```

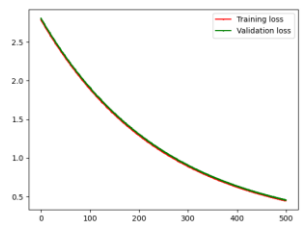
c. Dataset D




```
=== Linear Regression Evaluation | Dataset D | lr=0.1 ===
MSE       : 0.0824
MAE       : 0.2467
RMSE      : 0.2870
R-squared : 0.9716
```




```
=== Linear Regression Evaluation | Dataset D | lr=0.01 ===
MSE       : 0.0824
MAE       : 0.2467
RMSE      : 0.2870
R-squared : 0.9716
```




```
=== Linear Regression Evaluation | Dataset D | lr=0.001 ===
MSE       : 0.4475
MAE       : 0.5415
RMSE      : 0.6690
R-squared : 0.8458
```

Across all datasets, a similar pattern is observed:

lr = 0.1 → fast but less stable

lr = 0.001 → stable but slow

lr = 0.01 → best balance

This consistency indicates that a moderate learning rate provides robust performance across different data distributions and dimensions.

2.4 Overall Discussion (Task 1)

The experiments across Datasets A, B, C, and D consistently demonstrate that the learning rate plays a critical role in gradient-based optimization.

When the learning rate is set to 0.1, the model converges rapidly during early iterations, as reflected in the steep decline of the loss curves. However, noticeable oscillations can be observed, indicating that the updates are too large and may overshoot the optimal solution. This behavior leads to instability and slightly inferior evaluation performance.

On the other hand, a learning rate of 0.001 results in smooth and stable convergence across all datasets. Nevertheless, the convergence speed is significantly slower, and the

model may not fully reach the optimal solution within the given number of iterations, leading to suboptimal performance.

In contrast, a learning rate of 0.01 consistently achieves a balance between convergence speed and stability. The loss curves show steady and smooth decreases without significant fluctuations, and the evaluation metrics (lower MSE and higher R^2) confirm its superior performance across all datasets.

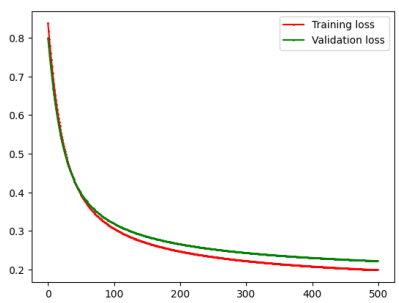
This consistent pattern across datasets with different characteristics suggests that a moderate learning rate provides robust and reliable optimization performance. Therefore, lr = 0.01 can be considered the most suitable choice for this task.

3. Task 2: Logistic Regression

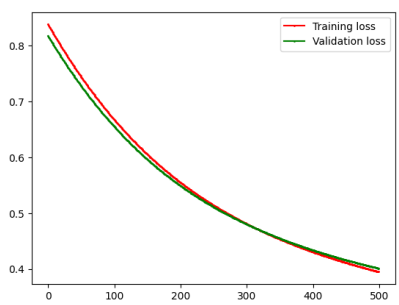
3.1 Learning Rate Experiment

To analyze classification performance, we evaluate logistic regression under different learning rates.

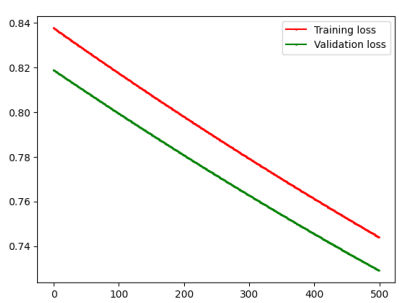
A. Dataset A




```
Logistic Regression Evaluation | Dataset A | lr=0.1
Accuracy  : 0.9050
Precision : 0.9087
Recall    : 0.9171
F1-score  : 0.9128
```




```
Logistic Regression Evaluation | Dataset A | lr=0.01
Accuracy  : 0.8775
Precision : 0.8784
Recall    : 0.8986
F1-score  : 0.8884
```



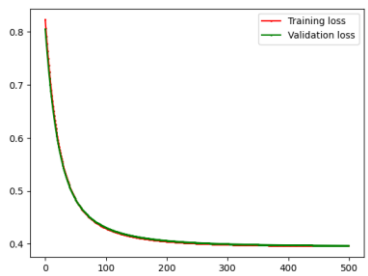

```
Logistic Regression Evaluation | Dataset A | lr=0.001
Accuracy  : 0.5325
Precision : 0.5676
Recall    : 0.5806
F1-score  : 0.5740
```

From these pictures, the learning rate of 0.1 reduces the loss quickly but shows fluctuations, indicating instability caused by large update steps. The learning rate of 0.001 converges smoothly but slowly. The learning rate of 0.01 provides stable convergence and achieves better classification performance.

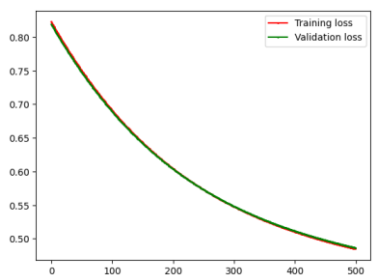
The evaluation metrics show that lr = 0.01 achieves higher Accuracy and F1-score, confirming its effectiveness.

B. Dataset B, C, D

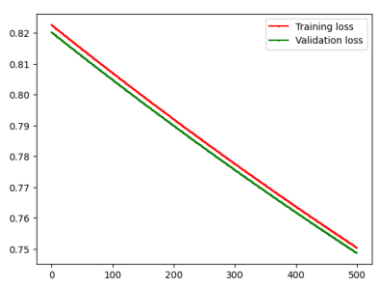
a. Dataset B




```
Logistic Regression Evaluation | Dataset B | lr=0.1
Accuracy  : 0.7800
Precision : 0.7793
Recall    : 0.8160
F1-score  : 0.7972
```

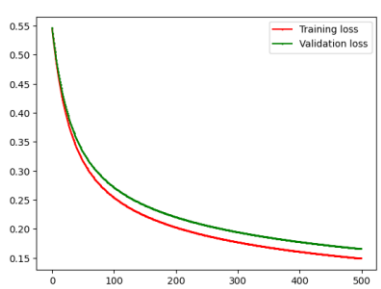



```
Logistic Regression Evaluation | Dataset B | lr=0.01
Accuracy  : 0.7725
Precision : 0.7689
Recall    : 0.8160
F1-score  : 0.7918
```

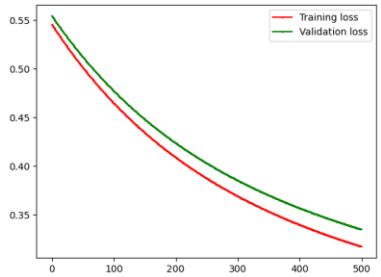



```
Logistic Regression Evaluation | Dataset B | lr=0.001
Accuracy  : 0.5275
Precision : 0.5516
Recall    : 0.5802
F1-score  : 0.5655
```

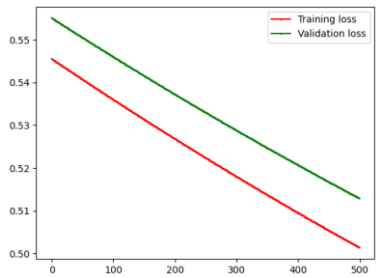
b. Dataset C




```
Logistic Regression Evaluation | Dataset C | lr=0.1
Accuracy  : 0.9762
Precision : 0.9852
Recall    : 0.9719
F1-score  : 0.9785
```

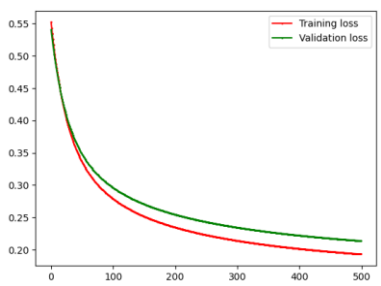



```
Logistic Regression Evaluation | Dataset C | lr=0.01
Accuracy  : 0.9150
Precision : 0.9353
Recall    : 0.9100
F1-score  : 0.9225
```

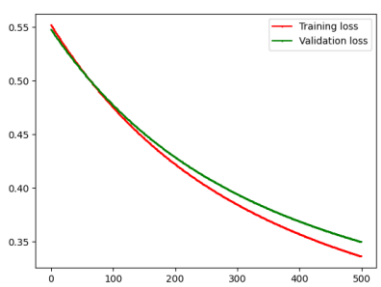



```
Logistic Regression Evaluation | Dataset C | lr=0.001
Accuracy  : 0.7250
Precision : 0.7761
Recall    : 0.7098
F1-score  : 0.7415
```

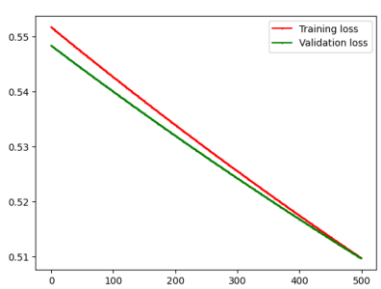
C. Dataset D




```
Logistic Regression Evaluation | Dataset D | lr=0.1
Accuracy  : 0.9225
Precision : 0.9330
Recall    : 0.9266
F1-score  : 0.9298
```




```
Logistic Regression Evaluation | Dataset D | lr=0.01
Accuracy  : 0.8775
Precision : 0.8984
Recall    : 0.8781
F1-score  : 0.8881
```



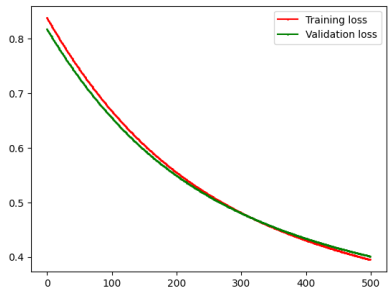

```
Logistic Regression Evaluation | Dataset D | lr=0.001
Accuracy  : 0.7169
Precision : 0.7670
Recall    : 0.7020
F1-score  : 0.7331
```

The same pattern is observed across all datasets, indicating that lr = 0.01 is consistently the most suitable learning rate.

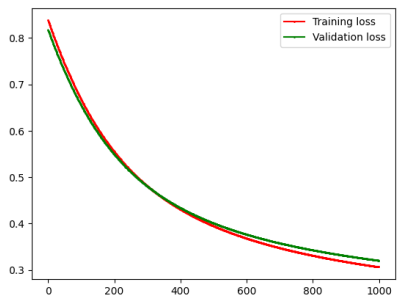
3.2 Iteration Experiment

To evaluate the effect of training duration, we fix the learning rate at 0.01 and vary the number of iterations.

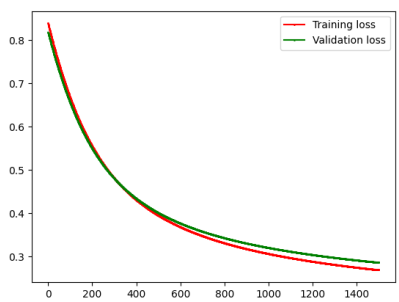
A. Dataset A




```
Logistic Regression Evaluation | Dataset A | n_iteration=500
Accuracy  : 0.8775
Precision : 0.8784
Recall    : 0.8986
F1-score  : 0.8884
```




```
Logistic Regression Evaluation | Dataset A | n_iteration=1000
Accuracy  : 0.9025
Precision : 0.9045
Recall    : 0.9171
F1-score  : 0.9108
```

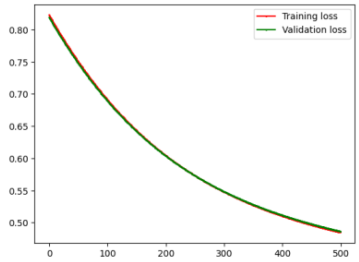



```
Logistic Regression Evaluation | Dataset A | n_iteration=1500
Accuracy  : 0.9052
Precision : 0.9083
Recall    : 0.9154
F1-score  : 0.9103
```

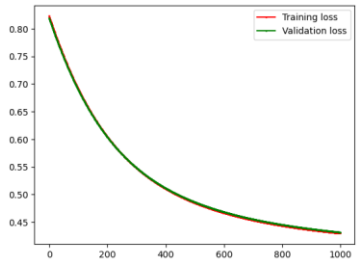
As shown in these pictures above, increasing iterations from 500 to 1000 significantly reduces the loss, indicating that the model continues to learn meaningful patterns. However, the improvement from 1000 to 1500 iterations is minimal. This suggests that the model has already approached convergence. The evaluation results reflect this behavior, where performance improvements diminish as the number of iterations increases.

B. Dataset B, C, D

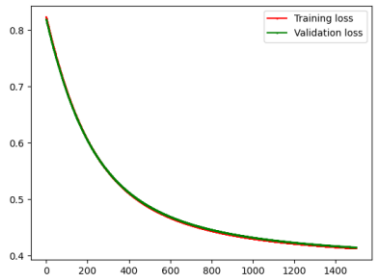
a. Dataset B




```
Logistic Regression Evaluation | Dataset B | n_iteration=500
Accuracy  : 0.7725
Precision : 0.7689
Recall    : 0.8160
F1-score  : 0.7918
```




```
Logistic Regression Evaluation | Dataset B | n_iteration=1000
Accuracy  : 0.7750
Precision : 0.7748
Recall    : 0.8113
F1-score  : 0.7926
```




```
Logistic Regression Evaluation | Dataset B | n_iteration=1500
Accuracy  : 0.7775
Precision : 0.7783
Recall    : 0.8113
F1-score  : 0.7945
```

The same trend is observed across all datasets, confirming the presence of diminishing returns in optimization.

3.3 Overall Discussion (Task 2)

The experiments on logistic regression demonstrate that both learning rate and the number of iterations play essential roles in model performance and convergence behavior.

From the learning rate experiments, a similar pattern to linear regression is observed across all datasets. A learning rate of 0.1 leads to rapid decreases in loss during early iterations, but noticeable fluctuations indicate instability caused by large update steps. In contrast, a learning rate of 0.001 produces smooth and stable convergence, but the learning process is significantly slower, and the model may not reach optimal performance within the given iterations. The learning rate of 0.01 consistently provides

the best balance, achieving stable convergence and superior evaluation performance, including higher Accuracy and F1-score.

The iteration experiments further reveal that increasing the number of iterations improves model performance up to a certain point. As the number of iterations increases from 500 to 1000, the loss decreases significantly, indicating that the model continues to learn meaningful patterns. However, the improvement from 1000 to 1500 iterations is minimal, suggesting that the model has already approached convergence and additional training provides diminishing returns.

These observations are consistent across all datasets, indicating that logistic regression performance depends on both efficient learning dynamics and sufficient training duration. Overall, the combination of a moderate learning rate (lr = 0.01) and an adequate number of iterations (approximately 1000) provides the most effective and stable performance.

4. Task 3: Data Preprocessing

4.1 Before Imputation

To understand the impact of missing values, we first compute the median and standard deviation of affected columns.

```
Columns with missing values:
SepalLengthCm     28
SepalWidthCm      57
PetalLengthCm    150
PetalWidthCm      22
BranchLength      24
dtype: int64
```

```
Median before imputation:
SepalLengthCm     6.300000
SepalWidthCm      2.900000
PetalLengthCm     5.085612
PetalWidthCm      1.600000
BranchLength     16.300000
dtype: float64
```

```
Standard deviation before imputation:
SepalLengthCm    1.037115
SepalWidthCm     0.389564
PetalLengthCm    1.582807
PetalWidthCm     0.706689
BranchLength     1.035228
dtype: float64
```

4.2 After KNN Imputation

We apply KNN imputation to fill missing values.

```
Comparison before vs after imputation:
               Median Before  Median After  Std Before  Std After
SepalLengthCm       6.300000      6.300000    1.037115   1.009275
SepalWidthCm        2.900000      2.900000    0.389564   0.372434
PetalLengthCm       5.085612      5.035683    1.582807   1.514955
PetalWidthCm        1.600000      1.700000    0.706689   0.694645
BranchLength       16.300000     16.300000    1.035228   1.011012
```

4.3 Analysis

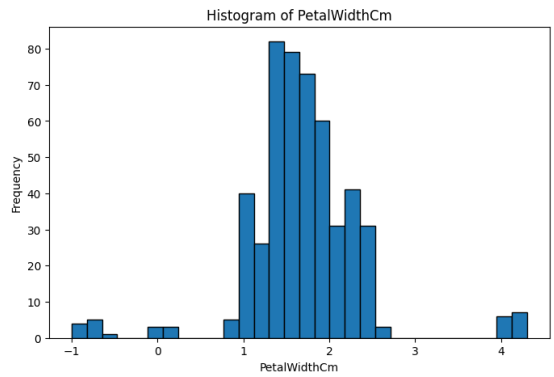
The median values remain relatively stable after imputation, indicating that the central tendency of the data is preserved. The standard deviation changes slightly, reflecting adjustments in data variability due to imputed values. KNN imputation

estimates missing values based on neighboring samples, preserving local data structure and producing more realistic distributions compared to simple imputation methods.

5. Task 4: Data Exploration

5.1 Histogram

To understand feature distribution, we visualize PetalWidthCm.



The histogram shows clustering patterns, suggesting potential separability between classes.

5.2 Correlation Analysis

We compute feature correlations with PetalWidthCm.

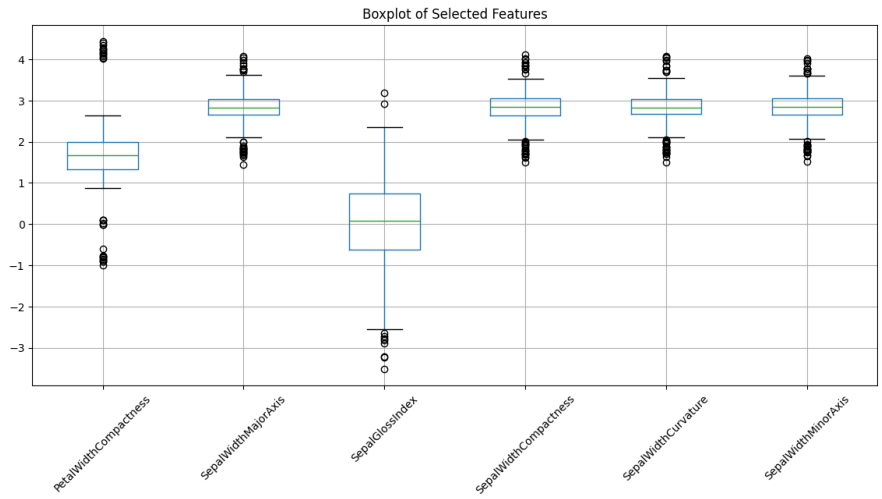
```
Largest positive correlation with PetalWidthCm:
PetalWidthCompactness: 0.991670

Top 5 strongest negative correlations with PetalWidthCm:
SepalWidthMajorAxis     -0.096350
SepalGlossIndex         -0.095200
SepalWidthCompactness   -0.088496
SepalWidthCurvature     -0.081334
SepalWidthMinorAxis     -0.074392
dtype: float64
```

The most positively correlated feature shows a strong direct relationship, while negatively correlated features indicate inverse relationships.

5.3 Boxplot

To analyze feature distributions, we visualize selected features.



The boxplot highlights differences in variability and distribution, providing insights into feature behavior.

5.4 Discussion

The data exploration results provide important insights into the structure and characteristics of the dataset. From the histogram of PetalWidthCm, clear clustering patterns can be observed, suggesting that the feature may help distinguish between different classes. This indicates that PetalWidthCm is a potentially informative feature for classification.

The correlation analysis further supports this observation. Features with strong positive correlation show a direct relationship with PetalWidthCm, while negatively correlated features indicate inverse relationships. These relationships highlight which features are more relevant and can contribute more effectively to the predictive model.

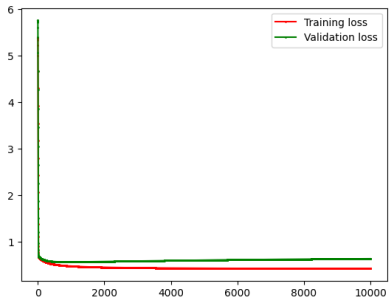
Additionally, the boxplots reveal differences in variability and distribution among features. Some features exhibit wider spreads and the presence of outliers, which may affect model stability and performance. Understanding these characteristics is important for feature selection and model robustness.

Overall, these analyses help identify informative features, reveal relationships between variables, and provide a better understanding of the dataset, which supports more effective model development.

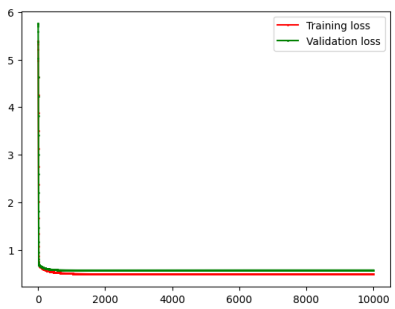
6. Task 5: Regularization

6.1 Loss Curve Comparison

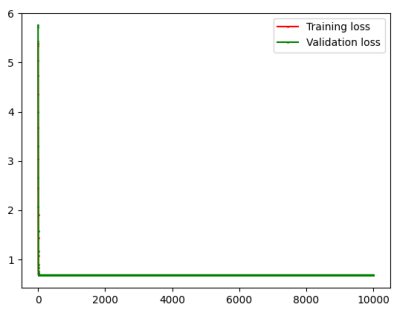
To analyze the effect of regularization, we compare different L2 penalty values.




```
Real-World Classification | L2 Regularization (lambda=0.01)
Accuracy  : 0.7200
Precision : 0.7125
Recall    : 0.7500
F1-score  : 0.7308
```




```
Real-World Classification | L2 Regularization (lambda=1)
Accuracy  : 0.7400
Precision : 0.7342
Recall    : 0.7632
F1-score  : 0.7484
```




```
Real-World Classification | L2 Regularization (lambda=100)
Accuracy  : 0.5067
Precision : 0.5067
Recall    : 1.0000
F1-score  : 0.6726
```

6.2 Analysis

As regularization strength increases, model parameters are constrained:

a. No regularization → risk of overfitting

b. Moderate λ → improved generalization

c. Large λ → underfitting

This illustrates the bias–variance trade-off.

7. Conclusion

This assignment demonstrates that model performance is strongly influenced by multiple components in the machine learning pipeline.

a. Linear Regression experiments, the learning rate of 0.01 consistently provides the best balance between convergence speed and stability across all datasets.
b. Logistic Regression, both learning rate and iteration are crucial. The best performance is achieved with a learning rate of 0.01 and approximately 1000 iterations, where the model converges effectively.
c. Real-world classification task, KNN imputation preserves the statistical properties of the dataset, while correlation analysis reveals that certain features have significantly stronger relationships with the target variable.
d. Regularization experiment demonstrates the bias–variance trade-off. Moderate L2 regularization improves generalization, while excessive regularization leads to underfitting.

Overall, careful tuning of hyperparameters, proper preprocessing, and appropriate regularization are essential for achieving optimal performance.